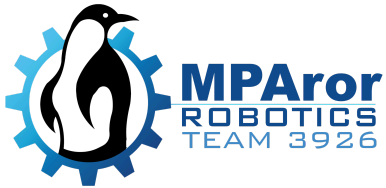
FRC Programming

Java foundations



A variable stores a value

A variable gives a name to information your program uses.

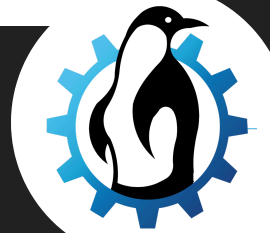
- `double speed = 0.5;` stores a decimal number.
- `boolean enabled = false;` stores a true or false value.
- A clear name explains what the value means.





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Methods



Values and decisions

Variables

A variable stores one piece of information.

- A speed request
- A sensor reading
- Whether a button is pressed

Conditions

An `if` statement chooses which code runs.

- Is the request positive?
- Has the value changed?
- Is the button pressed?



Live coding



Return values

A method can calculate a result and return it to the caller.

Here, `square(3.0)` returns `9.0`.

```
static double square(double value) {
    return value * value;
}

// Call the method.
double result = square(3.0);
```



Clear names make code easier to read



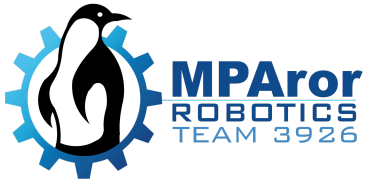
Practice

Reading Java

Predict each value before running the code.

Explain which expression produces the result.

Then change one input and compare your prediction.



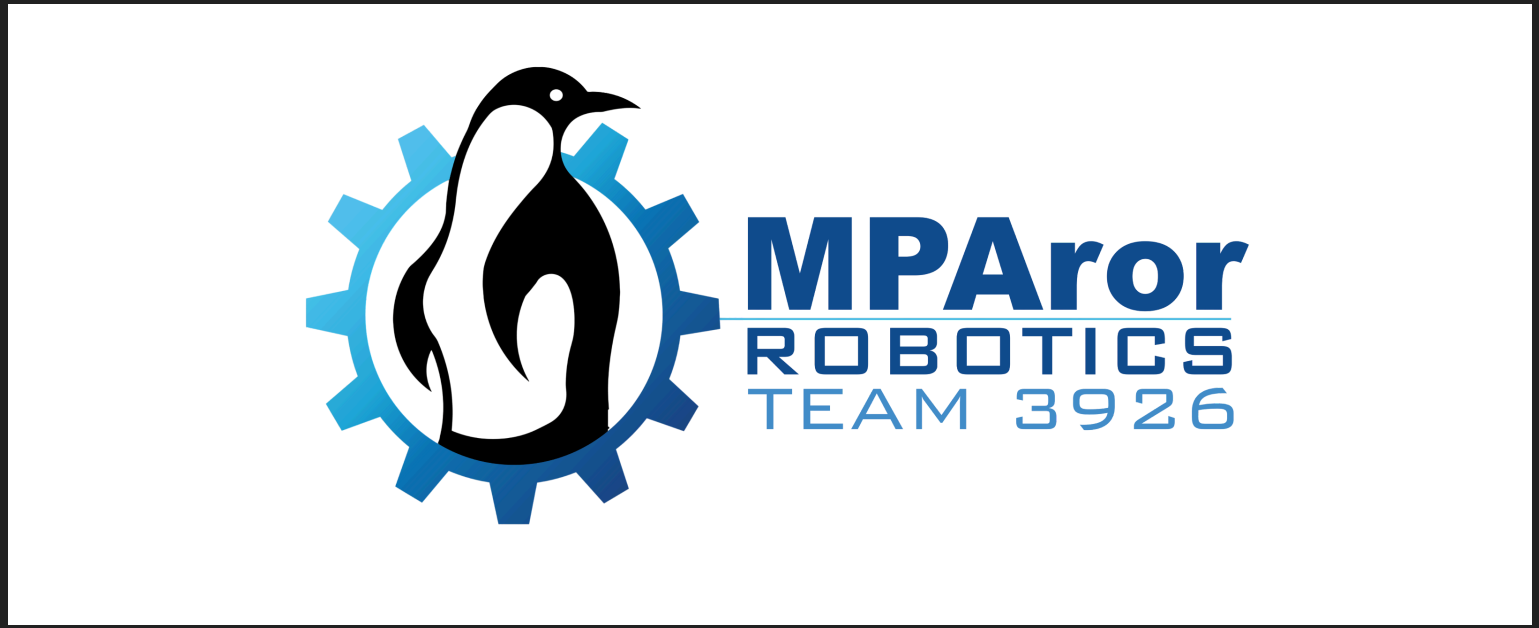
3926

MPArors Robotics





Our team



Replace this image with a robot photo, wiring diagram, or editor screenshot.

A method with a return value

```
static double clamp(double value) {
    if (value > 1.0) {
        return 1.0;
    }
    if (value < -1.0) {
        return -1.0;
    }
    return value;
}
```

This example keeps a value between −1.0 and 1.0.



Tracing an expression

Start with a request of `0.8`.

Multiplying by `0.5` gives `0.4`.

What happens if the request is negative?

```
// A calculation example.
double request = 0.8;
double scale = 0.5;
double output = request * scale;

System.out.println(output);
```

